

# Math is Finding Patterns

Lesson 1-5

**Name:**

**Date:**

**Class:**

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## My Notes

Level 1 Support



### TODAY'S OBJECTIVES

**Content Objective:** I can find patterns and pattern rules, use them to make generalizations, and check my solutions with a table of values.

**Language Objective:** I can describe a pattern using the words pattern rule, generalization, and 'each number is \_\_\_\_ more than the one before it.'



See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.



**Fill in each blank as we go. Use the Word Bank to help you.**



### WORD BANK — FILL EACH BLANK WITH THE BEST WORD

pattern

pattern rule

generalization

table of values

reasonableness



Tap any word to see what it means and a picture.

1

— Something that repeats or changes in a predictable way.

2

— The instruction that tells how to get from one number in a pattern to the next.

3

— A statement that is true for a whole pattern, made by noticing repeated calculations.

4

— A table that lists each step of a pattern so you can check that a solution is reasonable.

5

— Whether an answer makes sense — checked as you work, with adjustments made when needed.

## Write About the Math

Guided ESOL writing

Use the support level you need. Every level answers the same math question.

### 1. Understand the Question

JUSTIFY

**How can pattern rules tell you how many points each player has after the streak — and how can a table of values prove your answer is reasonable?**

**Your job:** Answer the question. Use math evidence. Explain how the evidence proves your answer.

*Responde la pregunta. Usa evidencia matemática. Explica cómo la evidencia demuestra tu respuesta.*

## 2. Plan Your Math Words

Check at least two words you will use.

☐ **pattern** — Something that repeats or changes in a predictable way.  
*patrón — Algo que se repite o cambia de una manera predecible.*

☐ **pattern rule** — The instruction that tells how to get from one number in a pattern to the next.  
*regla del patrón — La instrucción que dice cómo pasar de un número del patrón al siguiente.*

☐ **generalization** — A statement that is true for a whole pattern, made by noticing repeated calculations.  
*generalización — Una afirmación verdadera para todo un patrón, hecha al notar cálculos repetidos.*

☐ **table of values** — A table that lists each step of a pattern so you can check that a solution is reasonable.  
*tabla de valores — Una tabla que muestra cada paso de un patrón para comprobar que una solución es razonable.*

☐ **reasonableness** — Whether an answer makes sense — checked as you work, with adjustments made when needed.  
*sensatez del resultado — Si una respuesta tiene sentido — se comprueba mientras trabajas y se ajusta cuando es necesario.*

**Say it first:** Say your idea to a partner or quietly to yourself before you write.

*Di tu idea a un compañero o en voz baja antes de escribir.*

**Player A's pattern rule leads to a final score of \_\_\_\_ points. Player B's pattern rule leads to a final score of \_\_\_\_ points. Creating a \_\_\_\_ of values that checks each step confirms both totals are correct.**

*Di tu respuesta primero: Mi respuesta es \_\_\_\_ porque \_\_\_\_.*

### 3. Build Your Explanation

**Model the parts:** This model shows the parts of an explanation. It does not answer your writing question.

**Claim:** The lesson distinguishes patterns that keep going (like coin value scaling) from one-time facts (like the jar's specific total).

**Evidence:** A strong answer explains a pattern applies to any number of coins (it generalizes), while a fact like \$53.50 is fixed to this one jar. A weak answer just labels each card without explaining the difference.

**Reasoning:** This evidence connects the definition of pattern to the mathematical claim.

#### Start

Most language support

Complete the frames. Use the word bank.

*Completa las oraciones. Usa el banco de palabras.*

- **Player A's pattern rule leads to a final score of \_\_\_\_ points. Player B's pattern rule leads to a final score of \_\_\_\_ points. Creating a \_\_\_\_ of values that checks each step confirms both totals are correct.**
- **My answer is \_\_\_\_.**

*Mi respuesta es \_\_\_\_.*

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#### Build

Some language support

Write two connected sentences. Use because, so, or but.

*Escribe dos oraciones conectadas. Usa porque, entonces o pero.*

- **My claim is \_\_\_\_.**  
*Mi afirmación es \_\_\_\_.*
- **I know because \_\_\_\_.**  
*Lo sé porque \_\_\_\_.*
- **So \_\_\_\_.**  
*Entonces \_\_\_\_.*

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## Explain

### Light language support

Write a claim, give math evidence, and explain your reasoning.

*Escribe una afirmación, da evidencia matemática y explica tu razonamiento.*

- **My claim is \_\_\_\_.**  
*Mi afirmación es \_\_\_\_.*
  - **My math evidence is \_\_\_\_.**  
*Mi evidencia matemática es \_\_\_\_.*
  - **This proves \_\_\_\_ because \_\_\_\_.**  
*Esto demuestra \_\_\_\_ porque \_\_\_\_.*
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## 4. Check Your Explanation

- ☐ I answered the question.  
*Respondí la pregunta.*
- ☐ I used math evidence: numbers, an equation, a model, or a comparison.  
*Usé evidencia matemática: números, una ecuación, un modelo o una comparación.*
- ☐ I explained how the evidence proves my answer.  
*Explicué cómo la evidencia demuestra mi respuesta.*
- ☐ I used at least two math words.  
*Usé por lo menos dos palabras matemáticas.*
- ☐ I reread complete sentences and punctuation.  
*Volví a leer mis oraciones completas y la puntuación.*

